MIL-C-12095G
14 June 1982
SUPERSEDING
MIL-C-12095F
17 December 1971

MILITARY SPECIFICATION

CLOTH, WIND-RESISTANT SATEEN, COTTON; FIRE AND WATER RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of sateen cotton cloth which has been treated for flame and water resistance.

* 1.2 Classification. The cloth shall be of the following types (see 6.2):

Type I - Deleted (see 6.10)
Type II - 7.5 - 9.0 ounces per square yard

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
PPP-P-1134	- Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

* 3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

3.3 Material (see 6.11).

3.3.1 Yarn. The yarn shall be made from cotton which has been carded, combed, spun and twisted into either 2 or 3 ply yarns in the warp and filling (see 4.4.3 and 6.8).

3.4 Color. The color shall be as specified and shall match the standard sample (see 6.2, 6.3 and 6.5). The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as specified in 4.4.3. When no standard sample is available, the dyed cloth shall show no more than a slight trace of labile sulfur as defined in the test method when tested as specified in 4.4.3.

* 3.4.1 Matching. The color of the face side shall match the standard sample under artificial daylight having a color temperature of 7000 $\pm$ 500 kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 $\pm$ 100 kelvin.

3.4.2 Colorfastness. The finished cloth shall show colorfastness to weathering and crocking equal to or better than the standard sample when tested as specified in 4.4.3. When no standard sample is available or when the standard sample is not referenced for colorfastness, the finished cloth shall show "good" colorfastness to weathering and shall show no more crocking on dry white cloth than Munsell Value 7.0 when tested as specified in 4.4.3.

* 3.5 Physical requirements. The finished cloth shall conform to the physical requirements specified in table I when tested as specified in 4.4.3.

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TABLE I. Physical requirements

	Type II
Weight, (oz. per sq/yd.):	
Minimum	7.5
Maximum	9.0
Yarns per inch, (min.):	
Warp	104
Filling	88
Breaking strength, pounds (min.):	
Warp	170
Filling	150
Tearing strength, pounds (min.):	
Warp	6.0
Filling	6.0

* 3.5.1 Weave. The weave will be a 5-harness warp sateen with a counter of 2 as shown on figure 1. The use of fly-shuttle or a shuttleless loom is permitted.

* 3.5.2 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages. When the cloths are woven on shuttleless looms, the width measurement shall be made between the last warp yarn on each side, and the protruding fringe(s) are excluded.

3.5.3 Cloth preparation. The cloth shall be closely singed and mercerized. The cloth shall contain not more than 2.0 percent starch and protein content including chloroform-soluble and water-soluble material prior to water and fire resistant treatment when tested as specified in 4.4.3.

3.6 Finish. The cloth shall be given an approved finish (see 6.6).

* 3.6.1 Water repellent treatment. The cloth shall be given an approved durable water repellent finish. The finished cloth shall conform to the requirements specified in table II, and spray ratings of the three individual determinations of not less than 90, 90, 80 initially, and not less than 70, 70, 70 after 150 hours exposure to weathering when tested as specified in 4.4.3.

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TABLE II. Water repellency and air permeability requirements

	Type II
Hydrostatic pressure in centimeters minimum (average) 1/	45
Percent dynamic absorption (20-minute time interval) maximum (average)	25
Air permeability (cubic feet per minute per square foot) maximum (average)	3.0

1/ Low pressure range; no single determination shall fall below 40 cm.

* 3.6.2 Flame resistance. The cloth shall be given an approved durable flame-resistant treatment (see 6.6). The finished cloth shall have an after flame of not more than 2.0 seconds and an average char length of not more than 5.0 inches before and after three cycles of laundering when tested as specified in 4.4.3.

3.7 pH. The pH value of the water extract of finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.3.

* 3.8 Dimensional stability. The cloth shall not shrink or elongate more than 4.0 percent in either the direction of the warp or filling when tested as specified in 4.4.3.

3.9 Seam efficiency. The finished cloth shall have a seam efficiency of not less than 60 percent when tested as specified in 4.4.3.

3.10 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the piece.

3.11 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. The cloth shall be put-up on rolls as specified in PPP-P-1134.

3.12 Fiber identification. Each roll shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall conform to the quality established by this document. The demerit points per 100 square yards, when calculated as specified in section 4, shall not exceed the established maximum point values.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2.1 and 4.4.2.3 and tested for the characteristics specified in 4.4.3. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section 1 of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2.1.1 except that only those slubs and knots which exceed the limits shown on figure 1 of FED-STD-4 shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be 20 rolls selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 points. The lot shall be unacceptable if the

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points per 100 square yards of two or more individual rolls exceeds 45 points. If one roll exceeds 45 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 45 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Objectionable odor
Width less than specified
Cloudy, mottled, or streaky color
Not closely singed
Excessive neppiness

4.4.2.2 Length examination.

4.4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in a sample are defective in respect to length.

4.4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.2.3 Shade examination. During the yard-by-yard examination, each roll in the sample shall be examined for shade. Any roll in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

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4.4.2.4 Examination for compliance with Textile Fiber Products Identification Act and face marking. During the yard-by-yard examination, each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.
Face stamping missing from either or both ends.
Face stamping on wrong side.

* 4.4.3 End item testing. The cloth shall be tested for characteristics listed in table III. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 1/2 yard full width of the cloth prior to fire and water resistant treatments and 5 continuous yards full width of the finished cloth. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any of the requirements specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size in yards	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. End item tests

Characteristic	Requirement paragraph	Test method
Identification of cotton	3.3.1	1200 1/
Yarn ply (either 2 or 3 ply)	3.3.1	Visual 1/
Combed yarn	3.3.1	1/
Labile sulfur	3.4	2020
Colorfastness to:		
Crocking	3.4.2	5651
Weathering	3.4.2	5671 2/
Weight	3.5	5041
Yarns per inch	3.5	5050

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TABLE III. End item tests (cont'd)

Characteristic	Requirement paragraph	Test method
Breaking strength	3.5	5100
Tearing strength	3.5	5132
Weave	3.5.1	Visual 3/
Mercerization	3.5.3	1/
Starch and protein content including chloroform-soluble and water-soluble material	3.5.3	2611
Spray rating:		
Initial	3.6.1	5526
After weathering	3.6.1	5804, 5526 4/
Hydrostatic pressure	3.6.1	5514
Dynamic absorption	3.6.1	5500
Air permeability	3.6.1	5450
Flame resistance:		
After flame:		
Initial	3.6.2	5903
After laundering	3.6.2	5556, 5903
Char length:		
Initial	3.6.2	5903
After laundering	3.6.2	5556, 5903
pH	3.7	2811
Dimensional stability (type II)	3.8	5556
Seam efficiency	3.9	5110 5/

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ The time of exposure shall be 40 hours.

3/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

4/ The time of exposure shall be 150 hours.

5/ The thread shall conform to type IA1 of V-T-276. The needle size shall be 0.049 ± 0.001 inch, and the top (needle) thread shall be ticket No. 24, 3 ply and the bottom (looper) thread shall be ticket No. 50, 3 ply.

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4.4.4 Packaging inspection. Inspection shall be in accordance with the quality assurance provisions of PPP-P-1134.

5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging. Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. The cloth shall be put-up and packaged in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use. The cloth is intended for use in the fabrication of tentage.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) First article (see 3.2, 4.3 and 6.4).
(c) Color required (see 3.4).
(d) Width of cloth required (see 3.5.2).
(e) Length required, if other than specified (see 3.11).
(f) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.3 Standard sample. For access to standard sample address the contracting activity issuing the invitation for bids.

* 6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of 5 yards of the finished cloth. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Dyestuff formulation, Olive Green 107. A suggested but not mandatory dyestuff formulation for Olive Green 107 is as follows: The redness of shade shall result from the predominant use of the main colors specified and not from the shading colors:

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Shade Olive Green 107:

Vat Black 25, C.I. 69525
Vat Green 3, C.I. 69500
Vat Green 8, C.I. 71050
Shaded with Vat Yellow of
suitable fastness

6.6 Approval of finish. Approval of finish for utilization under this document is the responsibility of the US Army Natick Research and Development Laboratories, Natick, MA 01760 and is based on extensive tests, including those for toxicity which are not set forth in this specification. Because of the time necessary to conduct full evaluations, only those treatments approved by the appropriate medical service and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.7 Approval of flame resistants. As a part of the approval testing of new flame retardant finishes or fiber for tentage use, the finished cloth shall have a maximum of 2 seconds after flame and shall not melt or drop flaming pieces when tested in accordance with Method 5905 of FED-STD-191. The finished material shall retain flame resistance characteristic after 1 year of exposure to natural weathering.

6.8 Basic cloth. The cloth is based on commercial cloth made from 40/2 pima cotton for both warp and filling yarns.

* 6.9 Cloth treatment. The requirements specified for the cloth have been met using a combination of THPOH ammonia cure fire retardant finish and a buffered quarternary ammonium salt type water repellent.

* 6.10 Supersession data. Type I and it's requirements have been deleted. This type has been replaced by type II cloth. In end item documents where type I is specified, the requirements for type II cloth shall apply.

* 6.11 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.12 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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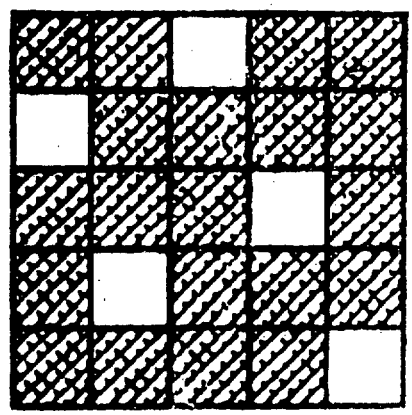


FIGURE 1. *Weave construction.*

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-12095G Cloth, Wind-Resistant Sateen, Cotton; Fire And Water Resistant

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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